Landscape II BA Semester Project

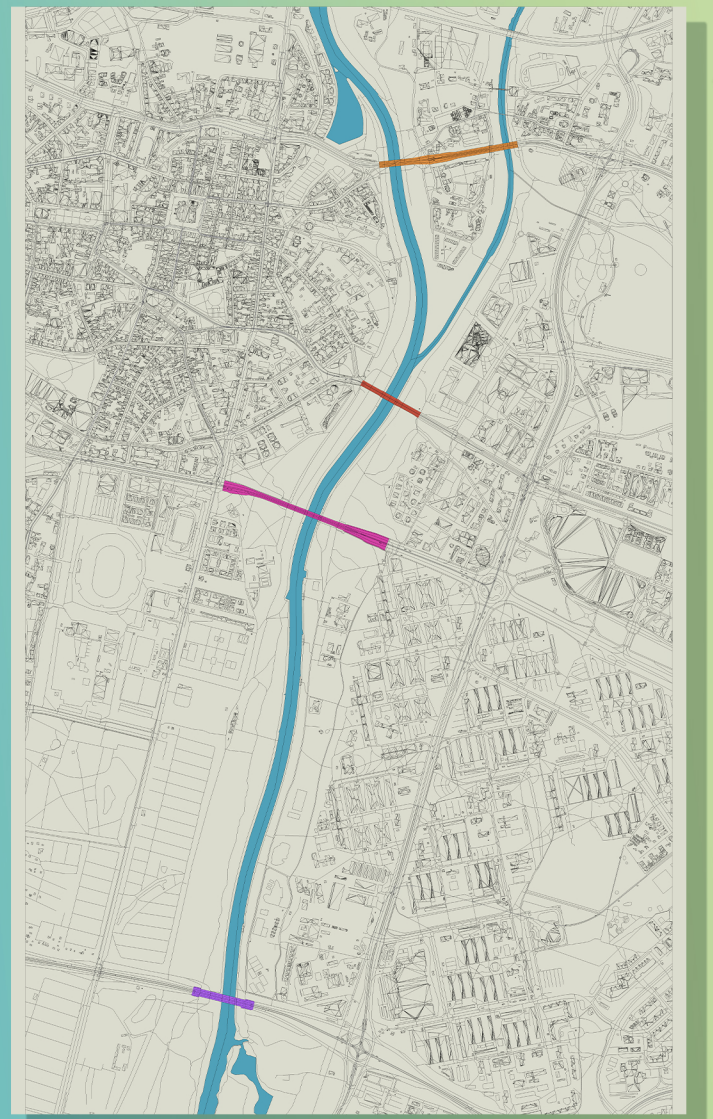
Landscape design project for an activity area located along the Warta River

Group Member
Zhang xinyue
Wang dingyi
Li zirui

This work was completed with the kind guidance of Dr. Eng. Izabela Myszkia.

UAP | POZNAŃ
Uniwersytet Artystyczny
im. Magdaleny Abakanowicz
w Poznaniu

located along the Warta River



The Warta River,
Poland's second-longest at 808.2 km,
flows northwest to join the Oder near
the German border. Its 54,529 km^2
basin is partly navigable from
Kostrzyn nad Odrą to Konin.

This semester's project focuses on
the area between several bridges
on the Warta River. Our goal is to
analyze the surroundings to develop
the best renovation plan and create
a vibrant space for community activities.

Legend

- Most Mieszka I Bridge
- Saint Roch Bridge
- Queen Jadwiga Bridge
- Przemysł I bridge
- Warta River

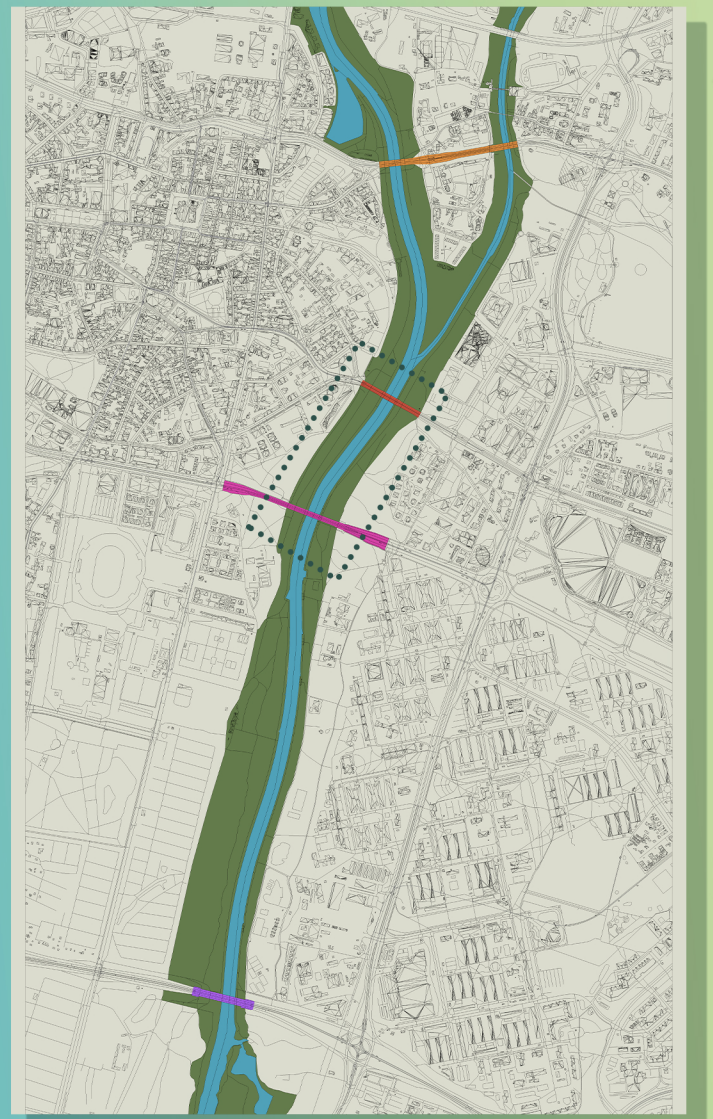
The Most Mieszka I Bridge in Poznań, Poland, spans the Warta River, connecting the city's east and west banks. Named after Poland's first ruler, it is a key transport hub with motor, bike, and pedestrian lanes. The bridge links residential areas, the business district near Stary Rynek, and educational institutions, while also offering access to scenic river landscapes and floodplain ecology.

The Saint Roch Bridge in central Poznan connects key areas across the Warta River:
Residential: Wilda District to the south offers mixed residential and commercial spaces.
Business: Stary Rynek (historical hub) and Swi ty Marcin Street (major commercial street) lie nearby.
Education: Adam Mickiewicz University (city center) and Poznan University of Technology (east bank) are close to the bridge.

The Queen Jadwiga Bridge in Poznań spans the Warta River, connecting dense residential areas, commercial zones, and educational institutions. It serves as a key traffic hub and symbolizes Poland's shift from communist to medieval heritage through its renaming.

The Przemysław I Bridge near the Warta River features extensive green spaces for recreation but limited development. It connects low-density residential areas with minimal commercial and educational facilities, serving mainly as a transportation route for vehicles, cyclists, and pedestrians.

The natural environment & vegetation



This is a small plain formed by the Varta River in this area
The flat areas on both sides of the Volta River were gradually formed by the river through erosion and sedimentation over a long geological period. The specific process includes:

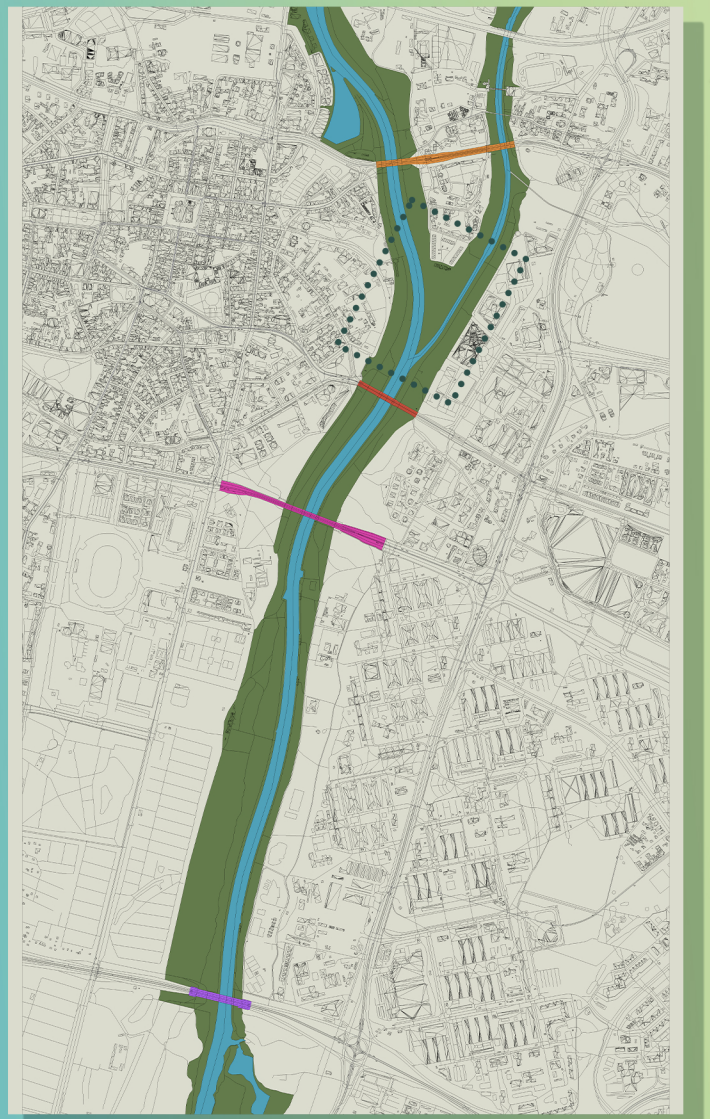
Water erosion: The water carried by rivers gradually erodes the riverbed and banks over time, especially in the bends and low slope areas of rivers, where erosion is most pronounced. The river continuously cuts through the surrounding soil and rocks, forming a wide valley.

The water flow pattern from upstream to downstream: In the upstream of the river, the water flow is faster, causing strong erosion; But downstream, the water flow is slower, the erosion force weakens, and sedimentation is more likely to occur.

Alluvial plain: When the Volta River flows through flat areas, it deposits sediment, gravel, and other debris on both sides of the river, forming an alluvial plain. These sediments are rich in organic matter and minerals, and the soil is fertile, which helps with vegetation growth.

Flood prone area: During the rainy season or spring when snow melts, the water level of the Volta River rises and floods occur. The

The natural environment & vegetation



Due to the presence of small wetlands in this area, the majority of vegetation is aquatic plants, which also provide habitat for the surrounding animal communities

This area is also a flat zone formed by erosion by flowing water, and has formed natural embankments Block the floods during the flood season, and due to the fact that this is a plain formed by river alluvial deposits, the soil fertility rate is high. Plants can be planted with a variety of plants to enrich the biodiversity here.

Poa annua L.

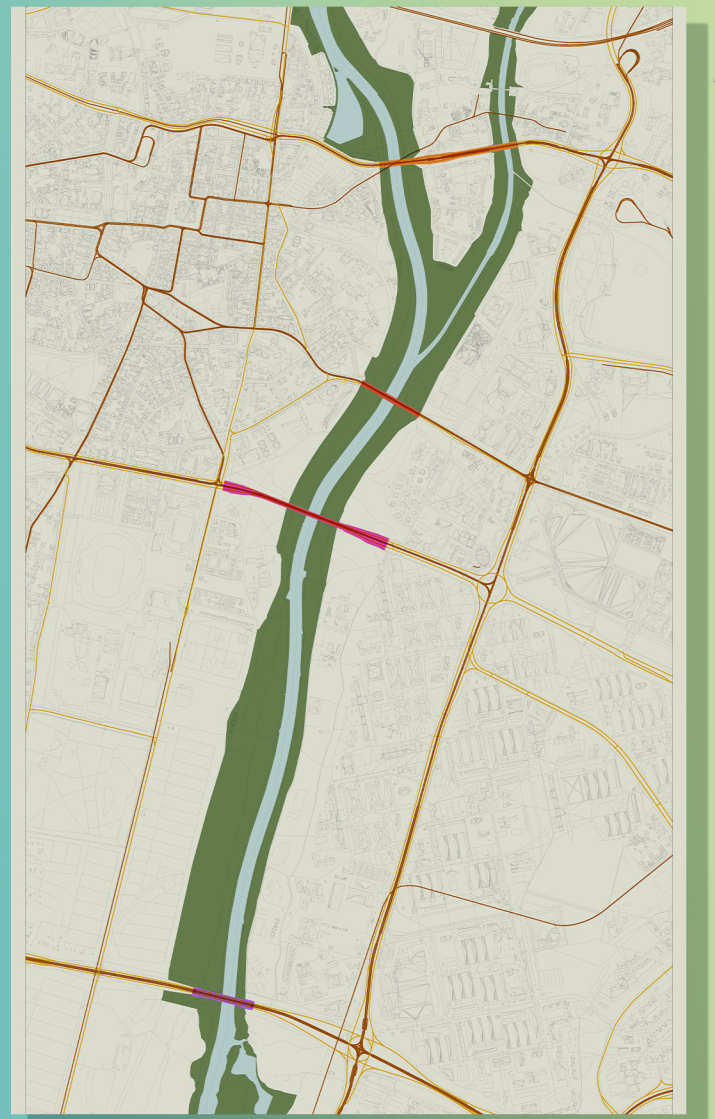
Reed

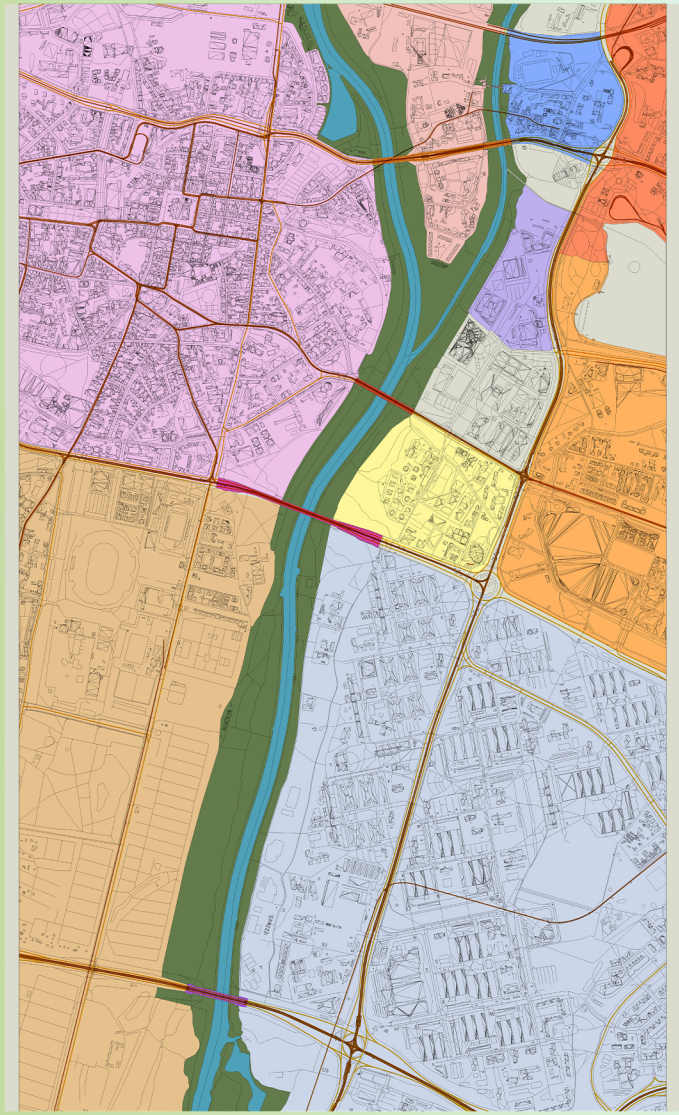
Sambucus williamsii

Moss

Juncus effusus L.

Solidago canadensis L.





Legend

- City Center
- Ostrow Tumski
- Srodka
- Komandoria
- Beedychowo
- Lacina
- S.W Rocha
- Rateje
- Wilda

1.3.1

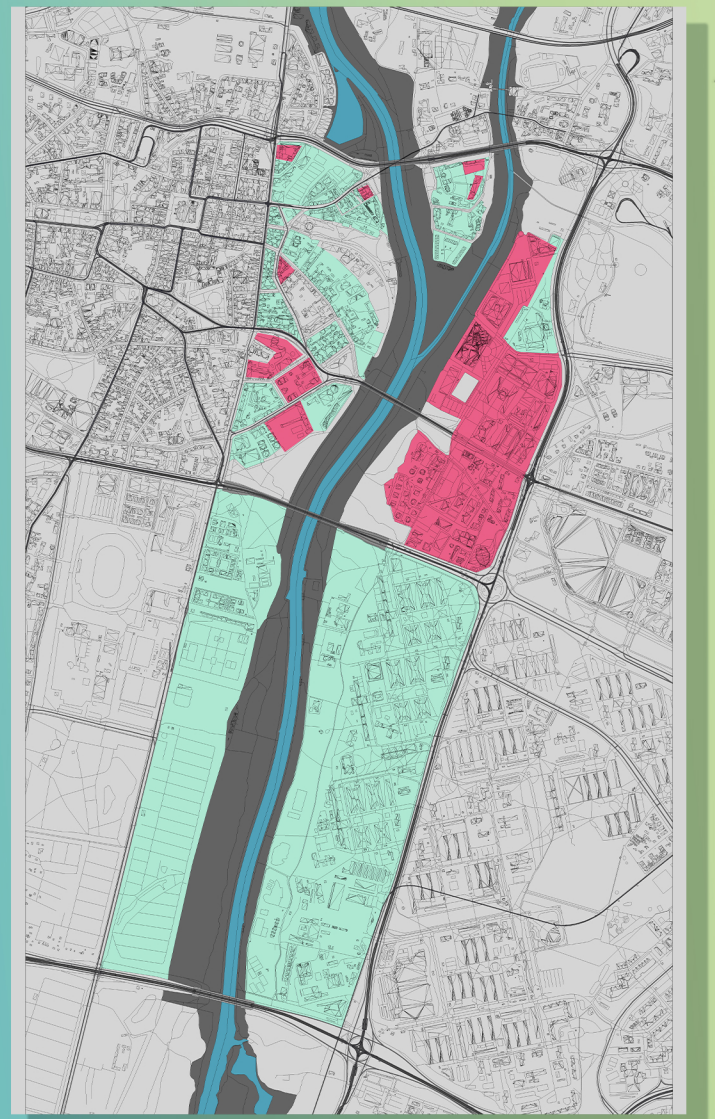
Transportation & residential areas

Transportation Division: Railways and main roads divide the city into distinct functional districts.
West Bank Focus: The city center and densely populated areas are concentrated on the west bank of the Warta River, serving as the economic and historical core.
East Bank Characteristics: The east bank is less developed, mainly residential, or reserved for future expansion.
Transportation Connectivity: The transportation network (railways and main roads) efficiently connects the districts while also acting as natural boundaries.
River's Role: The Warta River serves as a natural divider, with dense development on the west and more potential for growth on the east.

1.3.2

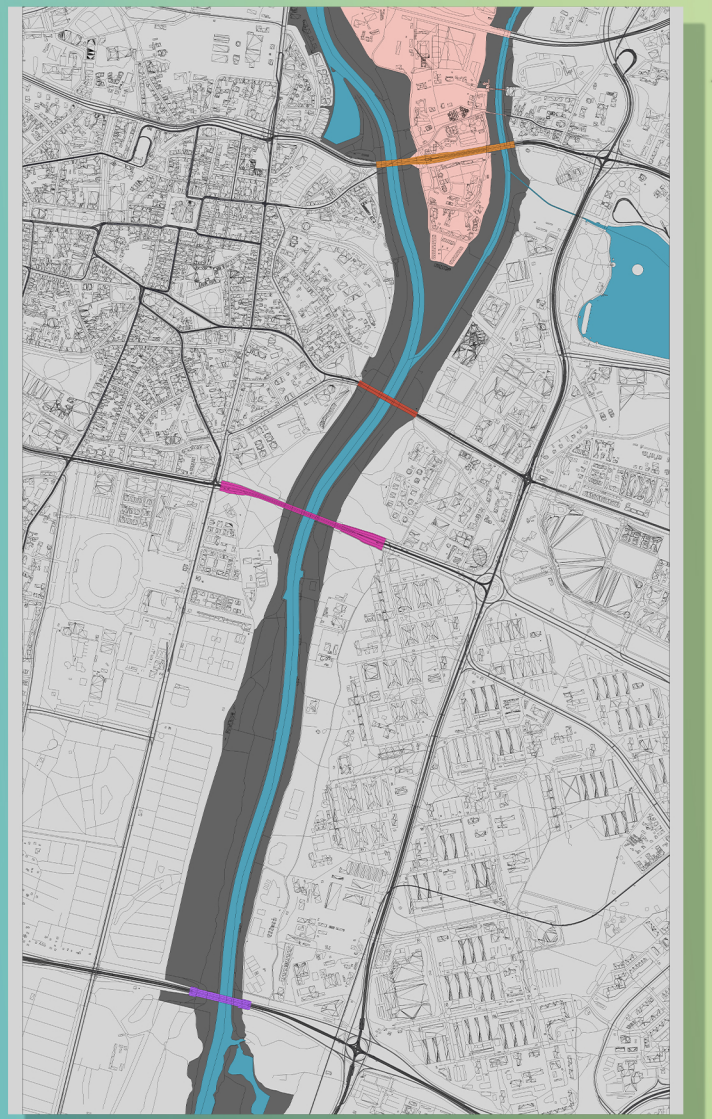
residential areas

Most people live in education areas and residential areas. The population structure of education areas is complex, with a higher proportion of foreign population and younger residential groups, while ordinary residential areas are mainly local citizens



- Companies cluster
- Open Space and parking lots
- Church and museum
- Natural landscapes and park
- Commercial district

Ostrow Tumski



Ostrow Tumski

Composition of Residential Population

• Religious figures : As this is an important center of the Polish Catholic Church, including Poznań Cathedral, many clergy, monks, and nuns reside in related institutions.
Mainly serving religious activities, church management, and tourist reception.
Cultural and academic personnel : There are academic institutions related to church and historical research in the neighborhood, such as the Institute of Religious History and Archives, which attract some researchers, scholars, and artists to live or work for a long time.

• A small number of ordinary residents : The number of ordinary residents living in this area is limited, mostly consisting of staff serving religious institutions, cultural sites, or tourism industries, as well as some local families who have lived here for a

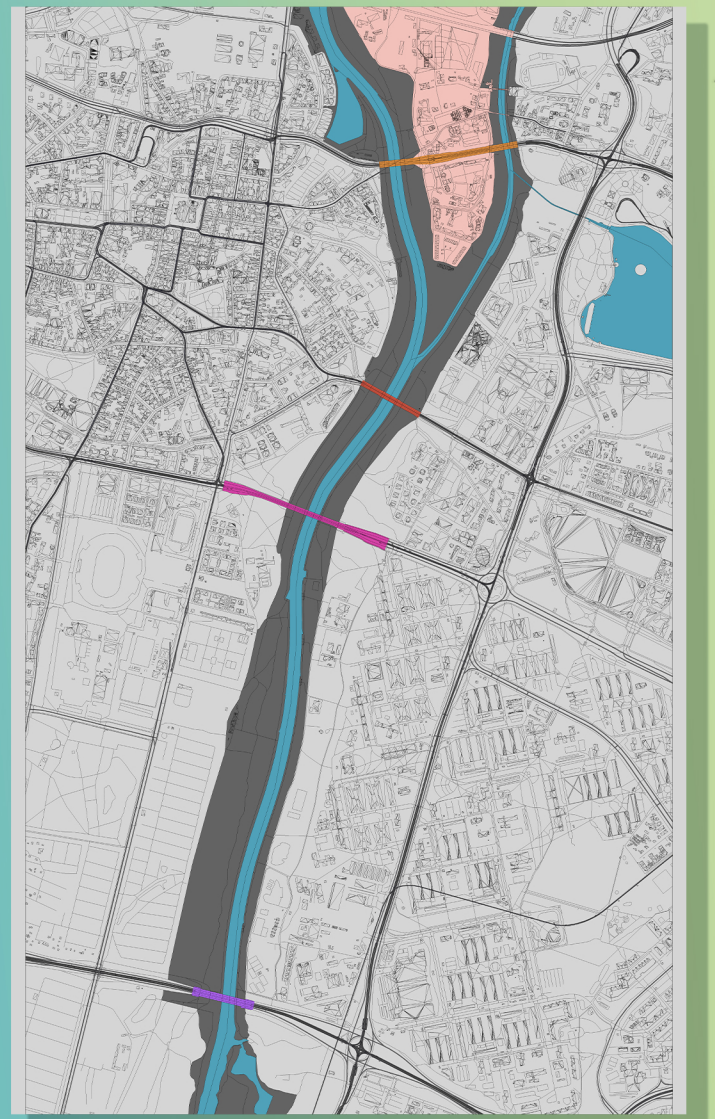
Floating population

• Tourists: As an important tourist landmark in Poznan, a large number of domestic and foreign tourists come to visit the cathedral, church museum, and surrounding historical sites.
The main audience is history and religious culture enthusiasts.

• Religious Pilgrims : Because the area is an important pilgrimage site for Catholics, it attracts many believers to regularly come here for religious activities.
The number and frequency of people are often related to important religious festivals or events such as Easter and Christmas.

• Students and scholars : The nearby religious and historical education institutions have attracted a certain number of students and researchers.

Ostrow Tumski



Ostrow Tumski

Population characteristics

- Age structure : Most clergy members are middle-aged and elderly people. Among the floating population, tourists and students are mostly young people.
- Occupational distribution : The majority are clergy and religious related personnel.Other industries include tourism services, cultural heritage management, academic research, etc.
- Ethnic and cultural background : Mainly consisting of Polish locals, there may be diverse cultural backgrounds among tourists and academic exchange personnel.

Detailed description

Number of permanent residents

- Estimated quantity: According to regional characteristics and living environment, the permanent population of Ostrów Tumski is approximately 200 to 300 people
- Restrictive factors: Due to the fact that most of Ostrów Tumski's area is for religious and public purposes (such as cathedrals, museums, green spaces, etc.), ordinary housing is very limited.

There are not many commercial facilities or modern communities within the block.

2. Number of floating population

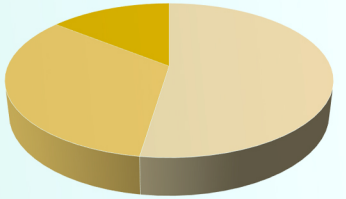
Daily floating population:

Tourists: As one of the most important tourist landmarks in Poznan, Ostrów Tumski attracts around 1000 to 3000 visitors daily, with even higher numbers during peak tourist seasons or religious holidays.

- Participants in religious activities: During major religious festivals or special religious events, the number of people attending Mass or pilgrimage may reach thousands.

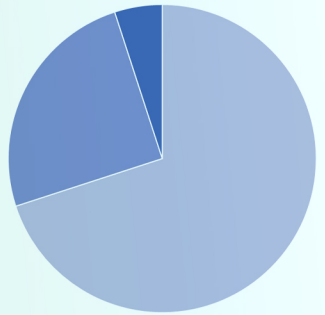
3. Long term trend: Due to the government's protection of cultural and religious values in the region, it will not be developed on a large scale, and there will be no significant changes in the permanent population

Composition of Residential Population



Religious figures
Cultural and academic personnel
ordinary residents

Age structure



Forty or over
18 - 39
Under 17